Lab1-Assignment

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This notebook describes the assignment for Lab 1 of the text mining course.

Points: each exercise is prefixed with the number of points you can obtain for the exercise.

We assume you have worked through the following notebooks:

- **Lab1.1-introduction**
- **Lab1.2-introduction-to-NLTK**
- **Lab1.3-introduction-to-spaCy**

In this assignment, you will process an English text (**Lab1-apple-samsung-example.txt**) with both NLTK and spaCy and discuss the similarities and differences.

Credits

The notebooks in this block have been originally created by Marten Postma. Adaptations were made by Filip Ilievski.

Tip: how to read a file from disk

Let's open the file **Lab1-apple-samsung-example.txt** from disk.

In [11]:
```
from pathlib import Path
```

In [12]:
```
cur_dir = Path().resolve() # this should provide you with the folder in which this notebook is placed
path_to_file = Path.joinpath(cur_dir, 'Lab1-apple-samsung-example.txt')
print(path_to_file)
print('does path exist? ->', Path.exists(path_to_file))
```

```
/Users/timmytimtim/Documents/College/Junior 2/Text Mining/text-mining-SP26/Labs/lab1/Lab1-apple-samsung-example.txt
does path exist? -> True
```

If the output from the code cell above states that **does path exist? -> False**, please check that the file **Lab1-apple-samsung-example.txt** is in the same directory as this notebook.

In [13]:
```
with open(path_to_file) as infile:
    text = infile.read()

print('number of characters', len(text))
```

```
number of characters 1139
```

[total points: 4] Exercise 1: NLTK

In this exercise, we use NLTK to apply **Part-of-speech (POS) tagging**, **Named Entity Recognition (NER)**, and **Constituency parsing**. The following code snippet already performs sentence splitting and tokenization.

In [14]:
```
import nltk
from nltk.tokenize import sent_tokenize
from nltk import word_tokenize
nltk.download('all')
```

```
[nltk_data] Downloading collection 'all'
[nltk_data]    |
[nltk_data]    | Downloading package abc to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package abc is already up-to-date!
[nltk_data]    | Downloading package alpino to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package alpino is already up-to-date!
[nltk_data]    | Downloading package averaged_perceptron_tagger to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package averaged_perceptron_tagger is already up-
[nltk_data]    |       to-date!
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[nltk_data]    | Downloading package basque_grammars to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
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[nltk_data]    | Downloading package bllip_wsj_no_aux to
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[nltk_data]    | Downloading package dependency_treebank to
```

```
[nltk_data]    |     /Users/timmytimtim/nltk_data...
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[nltk_data]    |   Package dolch is already up-to-date!
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[nltk_data]    | Downloading package maxent_treebank_pos_tagger to
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[nltk_data]    |       to-date!
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```

```
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[nltk_data]    | Downloading package punkt to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
```

```
[nltk_data]    |   Package punkt is already up-to-date!
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[nltk_data]    | Downloading package state_union to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package state_union is already up-to-date!
[nltk_data]    | Downloading package stopwords to
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[nltk_data]    |   Package stopwords is already up-to-date!
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[nltk_data]    |     /Users/timmytimtim/nltk_data...
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[nltk_data]    | Downloading package swadesh to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
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[nltk_data]    | Downloading package udhr to
```

```
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package udhr is already up-to-date!
[nltk_data]    | Downloading package udhr2 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package udhr2 is already up-to-date!
[nltk_data]    | Downloading package unicode_samples to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package unicode_samples is already up-to-date!
[nltk_data]    | Downloading package universal_tagset to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package universal_tagset is already up-to-date!
[nltk_data]    | Downloading package universal_treebanks_v20 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package universal_treebanks_v20 is already up-to-
[nltk_data]    |       date!
[nltk_data]    | Downloading package vader_lexicon to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package vader_lexicon is already up-to-date!
[nltk_data]    | Downloading package verbnet to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package verbnet is already up-to-date!
[nltk_data]    | Downloading package verbnet3 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package verbnet3 is already up-to-date!
[nltk_data]    | Downloading package webtext to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package webtext is already up-to-date!
[nltk_data]    | Downloading package wmt15_eval to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wmt15_eval is already up-to-date!
[nltk_data]    | Downloading package word2vec_sample to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package word2vec_sample is already up-to-date!
[nltk_data]    | Downloading package wordnet to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wordnet is already up-to-date!
[nltk_data]    | Downloading package wordnet2021 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wordnet2021 is already up-to-date!
[nltk_data]    | Downloading package wordnet2022 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wordnet2022 is already up-to-date!
[nltk_data]    | Downloading package wordnet31 to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wordnet31 is already up-to-date!
[nltk_data]    | Downloading package wordnet_ic to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package wordnet_ic is already up-to-date!
[nltk_data]    | Downloading package words to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package words is already up-to-date!
[nltk_data]    | Downloading package ycoe to
[nltk_data]    |     /Users/timmytimtim/nltk_data...
[nltk_data]    |   Package ycoe is already up-to-date!
[nltk_data]    |
[nltk_data]  Done downloading collection all
```

Out[14]: True

In [15]:
```
sentences_nltk = sent_tokenize(text)
```

In [16]:
```
tokens_per_sentence = []
for sentence_nltk in sentences_nltk:
    sent_tokens = word_tokenize(sentence_nltk)
    tokens_per_sentence.append(sent_tokens)
```

We will use lists to keep track of the output of the NLP tasks. We can hence inspect the output for each task using the index of the sentence.

In [17]:
```
sent_id = 1
print('SENTENCE', sentences_nltk[sent_id])
print('TOKENS', tokens_per_sentence[sent_id])
```

```
SENTENCE The six phones and tablets affected are the Galaxy S III, running the new Jelly Bean system, the Galaxy Tab 8.9 Wifi tablet, the Galaxy Tab 2 10.1, Galaxy Rugby Pro and Galaxy S III mini.
TOKENS ['The', 'six', 'phones', 'and', 'tablets', 'affected', 'are', 'the', 'Galaxy', 'S', 'III', ',', 'running', 'the', 'new', 'Jelly', 'Bean', 'system', ',', 'the', 'Galaxy', 'Tab', '8.9', 'Wifi', 'tablet', ',', 'the', 'Galaxy', 'Tab', '2', '10.1', ',', 'Galaxy', 'Rugby', 'Pro', 'and', 'Galaxy', 'S', 'III', 'mini', '.']
```

[point: 1] Exercise 1a: Part-of-speech (POS) tagging

Use `nltk.pos_tag` to perform part-of-speech tagging on each sentence.

Use `print` to **show** the output in the notebook (and hence also in the exported PDF!).

In [18]:
```
pos_tags_per_sentence = []
for tokens in tokens_per_sentence:
    curr_postag = nltk.pos_tag(tokens)
    pos_tags_per_sentence.append(curr_postag)
    print(curr_postag)
```

```
[('https', 'NN'), (':', ':'), ('//www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-produc
ts-under-scrutiny.html', 'JJ'), ('Documents', 'NNS'), ('filed', 'VBN'), ('to', 'TO'), ('the', 'DT'), ('San', 'NNP'),
('Jose', 'NNP'), ('federal', 'JJ'), ('court', 'NN'), ('in', 'IN'), ('California', 'NNP'), ('on', 'IN'), ('November',
'NNP'), ('23', 'CD'), ('list', 'NN'), ('six', 'CD'), ('Samsung', 'NNP'), ('products', 'NNS'), ('running', 'VBG'),
('the', 'DT'), ('``', '``'), ('Jelly', 'RB'), ('Bean', 'NNP'), ("''", "''"), ('and', 'CC'), ('``', '``'), ('Ice', 'N
NP'), ('Cream', 'NNP'), ('Sandwich', 'NNP'), ("''", "''"), ('operating', 'VBG'), ('systems', 'NNS'), (',', ','), ('w
hich', 'WDT'), ('Apple', 'NNP'), ('claims', 'VBZ'), ('infringe', 'VB'), ('its', 'PRP$'), ('patents', 'NNS'), ('.',
'.')]
[('The', 'DT'), ('six', 'CD'), ('phones', 'NNS'), ('and', 'CC'), ('tablets', 'NNS'), ('affected', 'VBN'), ('are', 'V
BP'), ('the', 'DT'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), (',', ','), ('running', 'VBG'), ('the', 'DT'),
('new', 'JJ'), ('Jelly', 'NNP'), ('Bean', 'NNP'), ('system', 'NN'), (',', ','), ('the', 'DT'), ('Galaxy', 'NNP'),
('Tab', 'NNP'), ('8.9', 'CD'), ('Wifi', 'NNP'), ('tablet', 'NN'), (',', ','), ('the', 'DT'), ('Galaxy', 'NNP'), ('Ta
b', 'NNP'), ('2', 'CD'), ('10.1', 'CD'), (',', ','), ('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro', 'NNP'), ('and', 'C
C'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), ('mini', 'NN'), ('.', '.')]
[('Apple', 'NNP'), ('stated', 'VBD'), ('it', 'PRP'), ('had', 'VBD'), ('“', 'NNP'), ('acted', 'VBD'), ('quickly', 'R
B'), ('and', 'CC'), ('diligently', 'RB'), ("''", "''"), ('in', 'IN'), ('order', 'NN'), ('to', 'TO'), ('``', '``'),
('determine', 'VB'), ('that', 'IN'), ('these', 'DT'), ('newly', 'RB'), ('released', 'VBN'), ('products', 'NNS'), ('d
o', 'VBP'), ('infringe', 'VB'), ('many', 'JJ'), ('of', 'IN'), ('the', 'DT'), ('same', 'JJ'), ('claims', 'NNS'), ('al
ready', 'RB'), ('asserted', 'VBN'), ('by', 'IN'), ('Apple', 'NNP'), ('.', '.'), ("''", "'')]
[('In', 'IN'), ('August', 'NNP'), (',', ','), ('Samsung', 'NNP'), ('lost', 'VBD'), ('a', 'DT'), ('US', 'NNP'), ('pat
ent', 'NN'), ('case', 'NN'), ('to', 'TO'), ('Apple', 'NNP'), ('and', 'CC'), ('was', 'VBD'), ('ordered', 'VBN'), ('t
o', 'TO'), ('pay', 'VB'), ('its', 'PRP$'), ('rival', 'JJ'), ('$', '$'), ('1.05bn', 'CD'), ('(', '('), ('£0.66bn', 'N
N'), (')', ')'), ('in', 'IN'), ('damages', 'NNS'), ('for', 'IN'), ('copying', 'VBG'), ('features', 'NNS'), ('of', 'I
N'), ('the', 'DT'), ('iPad', 'NN'), ('and', 'CC'), ('iPhone', 'NN'), ('in', 'IN'), ('its', 'PRP$'), ('Galaxy', 'NN
P'), ('range', 'NN'), ('of', 'IN'), ('devices', 'NNS'), ('.', '.')]
[('Samsung', 'NNP'), (',', ','), ('which', 'WDT'), ('is', 'VBZ'), ('the', 'DT'), ('world', 'NN'), ("'s", 'POS'), ('t
op', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN'), (',', ','), ('is', 'VBZ'), ('appealing', 'VBG'), ('t
he', 'DT'), ('ruling', 'NN'), ('.', '.')]
[('A', 'DT'), ('similar', 'JJ'), ('case', 'NN'), ('in', 'IN'), ('the', 'DT'), ('UK', 'NNP'), ('found', 'VBD'), ('i
n', 'IN'), ('Samsung', 'NNP'), ("'s", 'POS'), ('favour', 'NN'), ('and', 'CC'), ('ordered', 'VBD'), ('Apple', 'NNP'),
('to', 'TO'), ('publish', 'VB'), ('an', 'DT'), ('apology', 'NN'), ('making', 'VBG'), ('clear', 'JJ'), ('that', 'I
N'), ('the', 'DT'), ('South', 'JJ'), ('Korean', 'JJ'), ('firm', 'NN'), ('had', 'VBD'), ('not', 'RB'), ('copied', 'VB
N'), ('its', 'PRP$'), ('iPad', 'NN'), ('when', 'WRB'), ('designing', 'VBG'), ('its', 'PRP$'), ('own', 'JJ'), ('devic
es', 'NNS'), ('.', '.')]
```

In [19]:
```
print(pos_tags_per_sentence)
```

```
[[('https', 'NN'), (':', ':'), ('//www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-produ
cts-under-scrutiny.html', 'JJ'), ('Documents', 'NNS'), ('filed', 'VBN'), ('to', 'TO'), ('the', 'DT'), ('San', 'NN
P'), ('Jose', 'NNP'), ('federal', 'JJ'), ('court', 'NN'), ('in', 'IN'), ('California', 'NNP'), ('on', 'IN'), ('Novem
ber', 'NNP'), ('23', 'CD'), ('list', 'NN'), ('six', 'CD'), ('Samsung', 'NNP'), ('products', 'NNS'), ('running', 'VB
G'), ('the', 'DT'), ('``', '``'), ('Jelly', 'RB'), ('Bean', 'NNP'), ("''", "''"), ('and', 'CC'), ('``', '``'), ('Ic
e', 'NNP'), ('Cream', 'NNP'), ('Sandwich', 'NNP'), ("''", "''"), ('operating', 'VBG'), ('systems', 'NNS'), (',',
','), ('which', 'WDT'), ('Apple', 'NNP'), ('claims', 'VBZ'), ('infringe', 'VB'), ('its', 'PRP$'), ('patents', 'NN
S'), ('.', '.')], [('The', 'DT'), ('six', 'CD'), ('phones', 'NNS'), ('and', 'CC'), ('tablets', 'NNS'), ('affected',
'VBN'), ('are', 'VBP'), ('the', 'DT'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), (',', ','), ('running', 'VB
G'), ('the', 'DT'), ('new', 'JJ'), ('Jelly', 'NNP'), ('Bean', 'NNP'), ('system', 'NN'), (',', ','), ('the', 'DT'),
('Galaxy', 'NNP'), ('Tab', 'NNP'), ('8.9', 'CD'), ('Wifi', 'NNP'), ('tablet', 'NN'), (',', ','), ('the', 'DT'), ('Ga
laxy', 'NNP'), ('Tab', 'NNP'), ('2', 'CD'), ('10.1', 'CD'), (',', ','), ('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro',
'NNP'), ('and', 'CC'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), ('mini', 'NN'), ('.', '.')], [('Apple', 'NN
P'), ('stated', 'VBD'), ('it', 'PRP'), ('had', 'VBD'), ('“', 'NNP'), ('acted', 'VBD'), ('quickly', 'RB'), ('and', 'C
C'), ('diligently', 'RB'), ("''", "''"), ('in', 'IN'), ('order', 'NN'), ('to', 'TO'), ('``', '``'), ('determine', 'V
B'), ('that', 'IN'), ('these', 'DT'), ('newly', 'RB'), ('released', 'VBN'), ('products', 'NNS'), ('do', 'VBP'), ('in
fringe', 'VB'), ('many', 'JJ'), ('of', 'IN'), ('the', 'DT'), ('same', 'JJ'), ('claims', 'NNS'), ('already', 'RB'),
('asserted', 'VBN'), ('by', 'IN'), ('Apple', 'NNP'), ('.', '.'), ("''", "'')], [('In', 'IN'), ('August', 'NNP'),
(',', ','), ('Samsung', 'NNP'), ('lost', 'VBD'), ('a', 'DT'), ('US', 'NNP'), ('patent', 'NN'), ('case', 'NN'), ('t
o', 'TO'), ('Apple', 'NNP'), ('and', 'CC'), ('was', 'VBD'), ('ordered', 'VBN'), ('to', 'TO'), ('pay', 'VB'), ('its',
'PRP$'), ('rival', 'JJ'), ('$', '$'), ('1.05bn', 'CD'), ('(', '('), ('£0.66bn', 'NN'), (')', ')'), ('in', 'IN'), ('d
amages', 'NNS'), ('for', 'IN'), ('copying', 'VBG'), ('features', 'NNS'), ('of', 'IN'), ('the', 'DT'), ('iPad', 'N
N'), ('and', 'CC'), ('iPhone', 'NN'), ('in', 'IN'), ('its', 'PRP$'), ('Galaxy', 'NNP'), ('range', 'NN'), ('of', 'I
N'), ('devices', 'NNS'), ('.', '.')], [('Samsung', 'NNP'), (',', ','), ('which', 'WDT'), ('is', 'VBZ'), ('the', 'D
T'), ('world', 'NN'), ("'s", 'POS'), ('top', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN'), (',', ','),
('is', 'VBZ'), ('appealing', 'VBG'), ('the', 'DT'), ('ruling', 'NN'), ('.', '.')], [('A', 'DT'), ('similar', 'JJ'),
('case', 'NN'), ('in', 'IN'), ('the', 'DT'), ('UK', 'NNP'), ('found', 'VBD'), ('in', 'IN'), ('Samsung', 'NNP'),
("'s", 'POS'), ('favour', 'NN'), ('and', 'CC'), ('ordered', 'VBD'), ('Apple', 'NNP'), ('to', 'TO'), ('publish', 'V
B'), ('an', 'DT'), ('apology', 'NN'), ('making', 'VBG'), ('clear', 'JJ'), ('that', 'IN'), ('the', 'DT'), ('South',
'JJ'), ('Korean', 'JJ'), ('firm', 'NN'), ('had', 'VBD'), ('not', 'RB'), ('copied', 'VBN'), ('its', 'PRP$'), ('iPad',
'NN'), ('when', 'WRB'), ('designing', 'VBG'), ('its', 'PRP$'), ('own', 'JJ'), ('devices', 'NNS'), ('.', '.')]]
```

[point: 1] Exercise 1b: Named Entity Recognition (NER)

Use `nltk.chunk.ne_chunk` to perform Named Entity Recognition (NER) on each sentence.

Use `print` to **show** the output in the notebook (and hence also in the exported PDF!).

In [20]:
```python
from nltk.chunk import ne_chunk
ner_tags_per_sentence = []

for tokens_pos_tagged in pos_tags_per_sentence:
    tokens_pos_tagged_and_named_entities = ne_chunk(tokens_pos_tagged)
    ner_tags_per_sentence.append(tokens_pos_tagged_and_named_entities)
    print("")
    print('ORIGINAL SENTENCE', tokens_pos_tagged)
    print('NAMED ENTITY RECOGNITION OUTPUT', tokens_pos_tagged_and_named_entities)
```

```
ORIGINAL SENTENCE [('https', 'NN'), (':', ':'), ('//www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsu
it-six-more-products-under-scrutiny.html', 'JJ'), ('Documents', 'NNS'), ('filed', 'VBN'), ('to', 'TO'), ('the', 'D
T'), ('San', 'NNP'), ('Jose', 'NNP'), ('federal', 'JJ'), ('court', 'NN'), ('in', 'IN'), ('California', 'NNP'), ('o
n', 'IN'), ('November', 'NNP'), ('23', 'CD'), ('list', 'NN'), ('six', 'CD'), ('Samsung', 'NNP'), ('products', 'NN
S'), ('running', 'VBG'), ('the', 'DT'), ('``', '``'), ('Jelly', 'RB'), ('Bean', 'NNP'), ("''", "''"), ('and', 'CC'),
('``', '``'), ('Ice', 'NNP'), ('Cream', 'NNP'), ('Sandwich', 'NNP'), ("''", "''"), ('operating', 'VBG'), ('systems',
'NNS'), (',', ','), ('which', 'WDT'), ('Apple', 'NNP'), ('claims', 'VBZ'), ('infringe', 'VB'), ('its', 'PRP$'), ('pa
tents', 'NNS'), ('.', '.')]
NAMED ENTITY RECOGNITION OUTPUT (S
  https/NN
  :/:
  //www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html/JJ
  Documents/NNS
  filed/VBN
  to/TO
  the/DT
  (ORGANIZATION San/NNP Jose/NNP)
  federal/JJ
  court/NN
  in/IN
  (GPE California/NNP)
  on/IN
  November/NNP
  23/CD
  list/NN
  six/CD
  (ORGANIZATION Samsung/NNP)
  products/NNS
  running/VBG
  the/DT
  ``/``
  Jelly/RB
  (GPE Bean/NNP)
  ''/''
  and/CC
  ``/``
  Ice/NNP
  Cream/NNP
  Sandwich/NNP
  ''/''
  operating/VBG
  systems/NNS
  ,/,
  which/WDT
  (PERSON Apple/NNP)
  claims/VBZ
  infringe/VB
  its/PRP$
  patents/NNS
  ./.)

ORIGINAL SENTENCE [('The', 'DT'), ('six', 'CD'), ('phones', 'NNS'), ('and', 'CC'), ('tablets', 'NNS'), ('affected',
'VBN'), ('are', 'VBP'), ('the', 'DT'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), (',', ','), ('running', 'VB
G'), ('the', 'DT'), ('new', 'JJ'), ('Jelly', 'NNP'), ('Bean', 'NNP'), ('system', 'NN'), (',', ','), ('the', 'DT'),
('Galaxy', 'NNP'), ('Tab', 'NNP'), ('8.9', 'CD'), ('Wifi', 'NNP'), ('tablet', 'NN'), (',', ','), ('the', 'DT'), ('Ga
laxy', 'NNP'), ('Tab', 'NNP'), ('2', 'CD'), ('10.1', 'CD'), (',', ','), ('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro',
'NNP'), ('and', 'CC'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), ('mini', 'NN'), ('.', '.')]
NAMED ENTITY RECOGNITION OUTPUT (S
  The/DT
  six/CD
  phones/NNS
  and/CC
  tablets/NNS
  affected/VBN
  are/VBP
  the/DT
  (ORGANIZATION Galaxy/NNP)
  S/NNP
  III/NNP
  ,/,
  running/VBG
  the/DT
  new/JJ
  (PERSON Jelly/NNP Bean/NNP)
  system/NN
  ,/,
  the/DT
  (ORGANIZATION Galaxy/NNP)
  Tab/NNP
  8.9/CD
```

```
  Wifi/NNP
  tablet/NN
  ,/,
  the/DT
  (ORGANIZATION Galaxy/NNP)
  Tab/NNP
  2/CD
  10.1/CD
  ,/,
  (PERSON Galaxy/NNP Rugby/NNP Pro/NNP)
  and/CC
  (PERSON Galaxy/NNP S/NNP)
  III/NNP
  mini/NN
  ./.)

ORIGINAL SENTENCE [('Apple', 'NNP'), ('stated', 'VBD'), ('it', 'PRP'), ('had', 'VBD'), ('“', 'NNP'), ('acted', 'VB
D'), ('quickly', 'RB'), ('and', 'CC'), ('diligently', 'RB'), ("''", "''"), ('in', 'IN'), ('order', 'NN'), ('to', 'T
O'), ('``', '``'), ('determine', 'VB'), ('that', 'IN'), ('these', 'DT'), ('newly', 'RB'), ('released', 'VBN'), ('pro
ducts', 'NNS'), ('do', 'VBP'), ('infringe', 'VB'), ('many', 'JJ'), ('of', 'IN'), ('the', 'DT'), ('same', 'JJ'), ('cl
aims', 'NNS'), ('already', 'RB'), ('asserted', 'VBN'), ('by', 'IN'), ('Apple', 'NNP'), ('.', '.'), ("''", "''")]
NAMED ENTITY RECOGNITION OUTPUT (S
  (PERSON Apple/NNP)
  stated/VBD
  it/PRP
  had/VBD
  “/NNP
  acted/VBD
  quickly/RB
  and/CC
  diligently/RB
  ''/''
  in/IN
  order/NN
  to/TO
  ``/``
  determine/VB
  that/IN
  these/DT
  newly/RB
  released/VBN
  products/NNS
  do/VBP
  infringe/VB
  many/JJ
  of/IN
  the/DT
  same/JJ
  claims/NNS
  already/RB
  asserted/VBN
  by/IN
  (PERSON Apple/NNP)
  ./.
  ''/'')

ORIGINAL SENTENCE [('In', 'IN'), ('August', 'NNP'), (',', ','), ('Samsung', 'NNP'), ('lost', 'VBD'), ('a', 'DT'),
('US', 'NNP'), ('patent', 'NN'), ('case', 'NN'), ('to', 'TO'), ('Apple', 'NNP'), ('and', 'CC'), ('was', 'VBD'), ('or
dered', 'VBN'), ('to', 'TO'), ('pay', 'VB'), ('its', 'PRP$'), ('rival', 'JJ'), ('$', '$'), ('1.05bn', 'CD'), ('(',
'('), ('£0.66bn', 'NN'), (')', ')'), ('in', 'IN'), ('damages', 'NNS'), ('for', 'IN'), ('copying', 'VBG'), ('feature
s', 'NNS'), ('of', 'IN'), ('the', 'DT'), ('iPad', 'NN'), ('and', 'CC'), ('iPhone', 'NN'), ('in', 'IN'), ('its', 'PRP
$'), ('Galaxy', 'NNP'), ('range', 'NN'), ('of', 'IN'), ('devices', 'NNS'), ('.', '.')]
NAMED ENTITY RECOGNITION OUTPUT (S
  In/IN
  (GPE August/NNP)
  ,/,
  (PERSON Samsung/NNP)
  lost/VBD
  a/DT
  (GSP US/NNP)
  patent/NN
  case/NN
  to/TO
  (GPE Apple/NNP)
  and/CC
  was/VBD
  ordered/VBN
  to/TO
  pay/VB
  its/PRP$
```

```
  rival/JJ
  $/$
  1.05bn/CD
  (/(
  £0.66bn/NN
  )/)
  in/IN
  damages/NNS
  for/IN
  copying/VBG
  features/NNS
  of/IN
  the/DT
  (ORGANIZATION iPad/NN)
  and/CC
  (ORGANIZATION iPhone/NN)
  in/IN
  its/PRP$
  (GPE Galaxy/NNP)
  range/NN
  of/IN
  devices/NNS
  ./.)

ORIGINAL SENTENCE [('Samsung', 'NNP'), (',', ','), ('which', 'WDT'), ('is', 'VBZ'), ('the', 'DT'), ('world', 'NN'),
("'s", 'POS'), ('top', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN'), (',', ','), ('is', 'VBZ'), ('appea
ling', 'VBG'), ('the', 'DT'), ('ruling', 'NN'), ('.', '.')]
NAMED ENTITY RECOGNITION OUTPUT (S
  (GPE Samsung/NNP)
  ,/,
  which/WDT
  is/VBZ
  the/DT
  world/NN
  's/POS
  top/JJ
  mobile/NN
  phone/NN
  maker/NN
  ,/,
  is/VBZ
  appealing/VBG
  the/DT
  ruling/NN
  ./.)

ORIGINAL SENTENCE [('A', 'DT'), ('similar', 'JJ'), ('case', 'NN'), ('in', 'IN'), ('the', 'DT'), ('UK', 'NNP'), ('fou
nd', 'VBD'), ('in', 'IN'), ('Samsung', 'NNP'), ("'s", 'POS'), ('favour', 'NN'), ('and', 'CC'), ('ordered', 'VBD'),
('Apple', 'NNP'), ('to', 'TO'), ('publish', 'VB'), ('an', 'DT'), ('apology', 'NN'), ('making', 'VBG'), ('clear', 'J
J'), ('that', 'IN'), ('the', 'DT'), ('South', 'JJ'), ('Korean', 'JJ'), ('firm', 'NN'), ('had', 'VBD'), ('not', 'R
B'), ('copied', 'VBN'), ('its', 'PRP$'), ('iPad', 'NN'), ('when', 'WRB'), ('designing', 'VBG'), ('its', 'PRP$'), ('o
wn', 'JJ'), ('devices', 'NNS'), ('.', '.')]
NAMED ENTITY RECOGNITION OUTPUT (S
  A/DT
  similar/JJ
  case/NN
  in/IN
  the/DT
  (ORGANIZATION UK/NNP)
  found/VBD
  in/IN
  (GPE Samsung/NNP)
  's/POS
  favour/NN
  and/CC
  ordered/VBD
  (PERSON Apple/NNP)
  to/TO
  publish/VB
  an/DT
  apology/NN
  making/VBG
  clear/JJ
  that/IN
  the/DT
  (LOCATION South/JJ Korean/JJ)
  firm/NN
  had/VBD
  not/RB
  copied/VBN
```

```
its/PRP$
iPad/NN
when/WRB
designing/VBG
its/PRP$
own/JJ
devices/NNS
./.)
```

In [21]:
```
print(ner_tags_per_sentence)
```

```
[Tree('S', [('https', 'NN'), (':', ':'), ('//www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-
more-products-under-scrutiny.html', 'JJ'), ('Documents', 'NNS'), ('filed', 'VBN'), ('to', 'TO'), ('the', 'DT'), Tree
('ORGANIZATION', [('San', 'NNP'), ('Jose', 'NNP')]), ('federal', 'JJ'), ('court', 'NN'), ('in', 'IN'), Tree('GPE',
[('California', 'NNP')]), ('on', 'IN'), ('November', 'NNP'), ('23', 'CD'), ('list', 'NN'), ('six', 'CD'), Tree('ORGA
NIZATION', [('Samsung', 'NNP')]), ('products', 'NNS'), ('running', 'VBG'), ('the', 'DT'), ('``', '``'), ('Jelly', 'R
B'), Tree('GPE', [('Bean', 'NNP')]), ("''", "''"), ('and', 'CC'), ('``', '``'), ('Ice', 'NNP'), ('Cream', 'NNP'),
('Sandwich', 'NNP'), ("''", "''"), ('operating', 'VBG'), ('systems', 'NNS'), (',', ','), ('which', 'WDT'), Tree('PER
SON', [('Apple', 'NNP')]), ('claims', 'VBZ'), ('infringe', 'VB'), ('its', 'PRP$'), ('patents', 'NNS'), ('.', '.')]),
Tree('S', [('The', 'DT'), ('six', 'CD'), ('phones', 'NNS'), ('and', 'CC'), ('tablets', 'NNS'), ('affected', 'VBN'),
('are', 'VBP'), ('the', 'DT'), Tree('ORGANIZATION', [('Galaxy', 'NNP')]), ('S', 'NNP'), ('III', 'NNP'), (',', ','),
('running', 'VBG'), ('the', 'DT'), ('new', 'JJ'), Tree('PERSON', [('Jelly', 'NNP'), ('Bean', 'NNP')]), ('system', 'N
N'), (',', ','), ('the', 'DT'), Tree('ORGANIZATION', [('Galaxy', 'NNP')]), ('Tab', 'NNP'), ('8.9', 'CD'), ('Wifi',
'NNP'), ('tablet', 'NN'), (',', ','), ('the', 'DT'), Tree('ORGANIZATION', [('Galaxy', 'NNP')]), ('Tab', 'NNP'),
('2', 'CD'), ('10.1', 'CD'), (',', ','), Tree('PERSON', [('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro', 'NNP')]), ('an
d', 'CC'), Tree('PERSON', [('Galaxy', 'NNP'), ('S', 'NNP')]), ('III', 'NNP'), ('mini', 'NN'), ('.', '.')]), Tree
('S', [Tree('PERSON', [('Apple', 'NNP')]), ('stated', 'VBD'), ('it', 'PRP'), ('had', 'VBD'), ('“', 'NNP'), ('acted',
'VBD'), ('quickly', 'RB'), ('and', 'CC'), ('diligently', 'RB'), ("''", "''"), ('in', 'IN'), ('order', 'NN'), ('to',
'TO'), ('``', '``'), ('determine', 'VB'), ('that', 'IN'), ('these', 'DT'), ('newly', 'RB'), ('released', 'VBN'), ('p
roducts', 'NNS'), ('do', 'VBP'), ('infringe', 'VB'), ('many', 'JJ'), ('of', 'IN'), ('the', 'DT'), ('same', 'JJ'),
('claims', 'NNS'), ('already', 'RB'), ('asserted', 'VBN'), ('by', 'IN'), Tree('PERSON', [('Apple', 'NNP')]), ('.',
'.'), ("''", "''")]), Tree('S', [('In', 'IN'), Tree('GPE', [('August', 'NNP')]), (',', ','), Tree('PERSON', [('Samsu
ng', 'NNP')]), ('lost', 'VBD'), ('a', 'DT'), Tree('GSP', [('US', 'NNP')]), ('patent', 'NN'), ('case', 'NN'), ('to',
'TO'), Tree('GPE', [('Apple', 'NNP')]), ('and', 'CC'), ('was', 'VBD'), ('ordered', 'VBN'), ('to', 'TO'), ('pay', 'V
B'), ('its', 'PRP$'), ('rival', 'JJ'), ('$', '$'), ('1.05bn', 'CD'), ('(', '('), ('£0.66bn', 'NN'), (')', ')'), ('i
n', 'IN'), ('damages', 'NNS'), ('for', 'IN'), ('copying', 'VBG'), ('features', 'NNS'), ('of', 'IN'), ('the', 'DT'),
Tree('ORGANIZATION', [('iPad', 'NN')]), ('and', 'CC'), Tree('ORGANIZATION', [('iPhone', 'NN')]), ('in', 'IN'), ('it
s', 'PRP$'), Tree('GPE', [('Galaxy', 'NNP')]), ('range', 'NN'), ('of', 'IN'), ('devices', 'NNS'), ('.', '.')]), Tree
('S', [Tree('GPE', [('Samsung', 'NNP')]), (',', ','), ('which', 'WDT'), ('is', 'VBZ'), ('the', 'DT'), ('world', 'N
N'), ("'s", 'POS'), ('top', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN'), (',', ','), ('is', 'VBZ'),
('appealing', 'VBG'), ('the', 'DT'), ('ruling', 'NN'), ('.', '.')]), Tree('S', [('A', 'DT'), ('similar', 'JJ'), ('ca
se', 'NN'), ('in', 'IN'), ('the', 'DT'), Tree('ORGANIZATION', [('UK', 'NNP')]), ('found', 'VBD'), ('in', 'IN'), Tree
('GPE', [('Samsung', 'NNP')]), ("'s", 'POS'), ('favour', 'NN'), ('and', 'CC'), ('ordered', 'VBD'), Tree('PERSON',
[('Apple', 'NNP')]), ('to', 'TO'), ('publish', 'VB'), ('an', 'DT'), ('apology', 'NN'), ('making', 'VBG'), ('clear',
'JJ'), ('that', 'IN'), ('the', 'DT'), Tree('LOCATION', [('South', 'JJ'), ('Korean', 'JJ')]), ('firm', 'NN'), ('had',
'VBD'), ('not', 'RB'), ('copied', 'VBN'), ('its', 'PRP$'), ('iPad', 'NN'), ('when', 'WRB'), ('designing', 'VBG'),
('its', 'PRP$'), ('own', 'JJ'), ('devices', 'NNS'), ('.', '.')])]
```

[points: 2] Exercise 1c: Constituency parsing

Use the `nltk.RegexpParser` to perform constituency parsing on each sentence.

Use `print` to **show** the output in the notebook (and hence also in the exported PDF!).

In [22]:
```
constituent_parser = nltk.RegexpParser('''
NP: {<DT>? <JJ>* <NN>*} # NP
P: {<IN>}           # Preposition
V: {<V.*>}          # Verb
PP: {<P> <NP>}      # PP -> P NP
VP: {<V> <NP|PP>*}  # VP -> V (NP|PP)*''')
```

In [23]:
```
constituency_output_per_sentence = []

for tokens_pos_tagged in pos_tags_per_sentence:
    constituent_structure = constituent_parser.parse(tokens_pos_tagged)
    constituency_output_per_sentence.append(constituent_structure)

for tree in constituency_output_per_sentence:
    print(tree) # print the sentence structure
```

```
(S
  (NP https/NN)
  :/:
  (NP
    //www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html/JJ)
  Documents/NNS
  (VP (V filed/VBN))
  to/TO
  (NP the/DT)
  San/NNP
  Jose/NNP
  (NP federal/JJ court/NN)
  (P in/IN)
  California/NNP
  (P on/IN)
  November/NNP
  23/CD
  (NP list/NN)
  six/CD
  Samsung/NNP
  products/NNS
  (VP (V running/VBG) (NP the/DT))
  ``/``
  Jelly/RB
  Bean/NNP
  ''/''
  and/CC
  ``/``
  Ice/NNP
  Cream/NNP
  Sandwich/NNP
  ''/''
  (VP (V operating/VBG))
  systems/NNS
  ,/,
  which/WDT
  Apple/NNP
  (VP (V claims/VBZ))
  (VP (V infringe/VB))
  its/PRP$
  patents/NNS
  ./.)
(S
  (NP The/DT)
  six/CD
  phones/NNS
  and/CC
  tablets/NNS
  (VP (V affected/VBN))
  (VP (V are/VBP) (NP the/DT))
  Galaxy/NNP
  S/NNP
  III/NNP
  ,/,
  (VP (V running/VBG) (NP the/DT new/JJ))
  Jelly/NNP
  Bean/NNP
  (NP system/NN)
  ,/,
  (NP the/DT)
  Galaxy/NNP
  Tab/NNP
  8.9/CD
  Wifi/NNP
  (NP tablet/NN)
  ,/,
  (NP the/DT)
  Galaxy/NNP
  Tab/NNP
  2/CD
  10.1/CD
  ,/,
  Galaxy/NNP
  Rugby/NNP
  Pro/NNP
  and/CC
  Galaxy/NNP
  S/NNP
  III/NNP
  (NP mini/NN)
```

```
  ./.)
(S
  Apple/NNP
  (VP (V stated/VBD))
  it/PRP
  (VP (V had/VBD))
  “/NNP
  (VP (V acted/VBD))
  quickly/RB
  and/CC
  diligently/RB
  ''/''
  (PP (P in/IN) (NP order/NN))
  to/TO
  ``/``
  (VP (V determine/VB) (PP (P that/IN) (NP these/DT)))
  newly/RB
  (VP (V released/VBN))
  products/NNS
  (VP (V do/VBP))
  (VP
    (V infringe/VB)
    (NP many/JJ)
    (PP (P of/IN) (NP the/DT same/JJ)))
  claims/NNS
  already/RB
  (VP (V asserted/VBN))
  (P by/IN)
  Apple/NNP
  ./.
  ''/'')
(S
  (P In/IN)
  August/NNP
  ,/,
  Samsung/NNP
  (VP (V lost/VBD) (NP a/DT))
  US/NNP
  (NP patent/NN case/NN)
  to/TO
  Apple/NNP
  and/CC
  (VP (V was/VBD))
  (VP (V ordered/VBN))
  to/TO
  (VP (V pay/VB))
  its/PRP$
  (NP rival/JJ)
  $/$
  1.05bn/CD
  (/(
  (NP £0.66bn/NN)
  )/)
  (P in/IN)
  damages/NNS
  (P for/IN)
  (VP (V copying/VBG))
  features/NNS
  (PP (P of/IN) (NP the/DT iPad/NN))
  and/CC
  (NP iPhone/NN)
  (P in/IN)
  its/PRP$
  Galaxy/NNP
  (NP range/NN)
  (P of/IN)
  devices/NNS
  ./.)
(S
  Samsung/NNP
  ,/,
  which/WDT
  (VP (V is/VBZ) (NP the/DT world/NN))
  's/POS
  (NP top/JJ mobile/NN phone/NN maker/NN)
  ,/,
  (VP (V is/VBZ))
  (VP (V appealing/VBG) (NP the/DT ruling/NN))
  ./.)
(S
```

```
  (NP A/DT similar/JJ case/NN)
  (PP (P in/IN) (NP the/DT))
  UK/NNP
  (VP (V found/VBD))
  (P in/IN)
  Samsung/NNP
  's/POS
  (NP favour/NN)
  and/CC
  (VP (V ordered/VBD))
  Apple/NNP
  to/TO
  (VP (V publish/VB) (NP an/DT apology/NN))
  (VP
    (V making/VBG)
    (NP clear/JJ)
    (PP (P that/IN) (NP the/DT South/JJ Korean/JJ firm/NN)))
  (VP (V had/VBD))
  not/RB
  (VP (V copied/VBN))
  its/PRP$
  (NP iPad/NN)
  when/WRB
  (VP (V designing/VBG))
  its/PRP$
  (NP own/JJ)
  devices/NNS
  ./.)
```

In [24]:
```
print(constituency_output_per_sentence)
```

```
[Tree('S', [Tree('NP', [('https', 'NN')]), (':', ':'), Tree('NP', [('//www.telegraph.co.uk/technology/apple/9702716/
Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html', 'JJ')]), ('Documents', 'NNS'), Tree('VP', [Tree('V',
[('filed', 'VBN')])]), ('to', 'TO'), Tree('NP', [('the', 'DT')]), ('San', 'NNP'), ('Jose', 'NNP'), Tree('NP', [('fed
eral', 'JJ'), ('court', 'NN')]), Tree('P', [('in', 'IN')]), ('California', 'NNP'), Tree('P', [('on', 'IN')]), ('Nove
mber', 'NNP'), ('23', 'CD'), Tree('NP', [('list', 'NN')]), ('six', 'CD'), ('Samsung', 'NNP'), ('products', 'NNS'), T
ree('VP', [Tree('V', [('running', 'VBG')]), Tree('NP', [('the', 'DT')])]), ('``', '``'), ('Jelly', 'RB'), ('Bean',
'NNP'), ("'", "'"), ('and', 'CC'), ('``', '``'), ('Ice', 'NNP'), ('Cream', 'NNP'), ('Sandwich', 'NNP'), ("'",
"'"), Tree('VP', [Tree('V', [('operating', 'VBG')])]), ('systems', 'NNS'), (',', ','), ('which', 'WDT'), ('Apple',
'NNP'), Tree('VP', [Tree('V', [('claims', 'VBZ')])]), Tree('VP', [Tree('V', [('infringe', 'VB')])])), ('its', 'PRP
$'), ('patents', 'NNS'), ('.', '.')]), Tree('S', [Tree('NP', [('The', 'DT')]), ('six', 'CD'), ('phones', 'NNS'), ('a
nd', 'CC'), ('tablets', 'NNS'), Tree('VP', [Tree('V', [('affected', 'VBN')])]), Tree('VP', [Tree('V', [('are', 'VB
P')]), Tree('NP', [('the', 'DT')])]), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), (',', ','), Tree('VP', [Tree
('V', [('running', 'VBG')]), Tree('NP', [('the', 'DT'), ('new', 'JJ')])]), ('Jelly', 'NNP'), ('Bean', 'NNP'), Tree
('NP', [('system', 'NN')]), (',', ','), Tree('NP', [('the', 'DT')]), ('Galaxy', 'NNP'), ('Tab', 'NNP'), ('8.9', 'C
D'), ('Wifi', 'NNP'), Tree('NP', [('tablet', 'NN')]), (',', ','), Tree('NP', [('the', 'DT')]), ('Galaxy', 'NNP'),
('Tab', 'NNP'), ('2', 'CD'), ('10.1', 'CD'), (',', ','), ('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro', 'NNP'), ('an
d', 'CC'), ('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP'), Tree('NP', [('mini', 'NN')]), ('.', '.')]), Tree('S',
[('Apple', 'NNP'), Tree('VP', [Tree('V', [('stated', 'VBD')])]), ('it', 'PRP'), Tree('VP', [Tree('V', [('had', 'VB
D')])]), ('“', 'NNP'), Tree('VP', [Tree('V', [('acted', 'VBD')])]), ('quickly', 'RB'), ('and', 'CC'), ('diligently',
'RB'), ("'", "'"), Tree('PP', [Tree('P', [('in', 'IN')]), Tree('NP', [('order', 'NN')])]), ('to', 'TO'), ('``', '`
`'), Tree('VP', [Tree('V', [('determine', 'VB')]), Tree('PP', [Tree('P', [('that', 'IN')]), Tree('NP', [('these', 'D
T')])])]), ('newly', 'RB'), Tree('VP', [Tree('V', [('released', 'VBN')])]), ('products', 'NNS'), Tree('VP', [Tree
('V', [('do', 'VBP')])]), Tree('VP', [Tree('V', [('infringe', 'VB')]), Tree('NP', [('many', 'JJ')]), Tree('PP', [Tre
e('P', [('of', 'IN')]), Tree('NP', [('the', 'DT'), ('same', 'JJ')])])]), ('claims', 'NNS'), ('already', 'RB'), Tree
('VP', [Tree('V', [('asserted', 'VBN')])])]), Tree('P', [('by', 'IN')]), ('Apple', 'NNP'), ('.', '.'), ("'", "'")]),
Tree('S', [Tree('P', [('In', 'IN')]), ('August', 'NNP'), (',', ','), ('Samsung', 'NNP'), Tree('VP', [Tree('V', [('lo
st', 'VBD')]), Tree('NP', [('a', 'DT')])]), ('US', 'NNP'), Tree('NP', [('patent', 'NN'), ('case', 'NN')]), ('to', 'T
O'), ('Apple', 'NNP'), ('and', 'CC'), Tree('VP', [Tree('V', [('was', 'VBD')])]), Tree('VP', [Tree('V', [('ordered',
'VBN')])]), ('to', 'TO'), Tree('VP', [Tree('V', [('pay', 'VB')])]), ('its', 'PRP$'), Tree('NP', [('rival', 'JJ')]),
('$', '$'), ('1.05bn', 'CD'), ('(', '('), Tree('NP', [('£0.66bn', 'NN')]), (')', ')'), Tree('P', [('in', 'IN')]),
('damages', 'NNS'), Tree('P', [('for', 'IN')]), Tree('VP', [Tree('V', [('copying', 'VBG')])]), ('features', 'NNS'),
Tree('PP', [Tree('P', [('of', 'IN')]), Tree('NP', [('the', 'DT'), ('iPad', 'NN')])]), ('and', 'CC'), Tree('NP', [('i
Phone', 'NN')]), Tree('P', [('in', 'IN')]), ('its', 'PRP$'), ('Galaxy', 'NNP'), Tree('NP', [('range', 'NN')]), Tree
('P', [('of', 'IN')]), ('devices', 'NNS'), ('.', '.')]), Tree('S', [('Samsung', 'NNP'), (',', ','), ('which', 'WD
T'), Tree('VP', [Tree('V', [('is', 'VBZ')]), Tree('NP', [('the', 'DT'), ('world', 'NN')])]), ("'s", 'POS'), Tree('N
P', [('top', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN')]), (',', ','), Tree('VP', [Tree('V', [('is',
'VBZ')])]), Tree('VP', [Tree('V', [('appealing', 'VBG')]), Tree('NP', [('the', 'DT'), ('ruling', 'NN')])]), ('.',
'.')]), Tree('S', [Tree('NP', [('A', 'DT'), ('similar', 'JJ'), ('case', 'NN')]), Tree('PP', [Tree('P', [('in', 'I
N')]), Tree('NP', [('the', 'DT')])]), ('UK', 'NNP'), Tree('VP', [Tree('V', [('found', 'VBD')])]), Tree('P', [('in',
'IN')]), ('Samsung', 'NNP'), ("'s", 'POS'), Tree('NP', [('favour', 'NN')]), ('and', 'CC'), Tree('VP', [Tree('V',
[('ordered', 'VBD')])]), ('Apple', 'NNP'), ('to', 'TO'), Tree('VP', [Tree('V', [('publish', 'VB')]), Tree('NP', [('a
n', 'DT'), ('apology', 'NN')])]), Tree('VP', [Tree('V', [('making', 'VBG')]), Tree('NP', [('clear', 'JJ')]), Tree('P
P', [Tree('P', [('that', 'IN')]), Tree('NP', [('the', 'DT'), ('South', 'JJ'), ('Korean', 'JJ'), ('firm', 'N
N')])])]), Tree('VP', [Tree('V', [('had', 'VBD')])]), ('not', 'RB'), Tree('VP', [Tree('V', [('copied', 'VBN')])]),
('its', 'PRP$'), Tree('NP', [('iPad', 'NN')]), ('when', 'WRB'), Tree('VP', [Tree('V', [('designing', 'VBG')])]), ('i
ts', 'PRP$'), Tree('NP', [('own', 'JJ')]), ('devices', 'NNS'), ('.', '.')])]
```

Augment the RegexpParser so that it also detects Named Entity Phrases (NEP), e.g., that it detects *Galaxy S III* and *Ice Cream Sandwich*

In [25]:
```
constituent_parser_v2 = nltk.RegexpParser('''
NEP: {<NNP>+}       # Named Entity Phrase -> one or more proper nouns
NP: {<DT>? <JJ>* <NN>*} # NP
P: {<IN>}           # Preposition
V: {<V.*>}          # Verb
PP: {<P> <NP>}      # PP -> P NP
VP: {<V> <NP|PP>*}  # VP -> V (NP|PP)*''')
```

In [26]:
```
constituency_v2_output_per_sentence = []
for tokens_pos_tagged in pos_tags_per_sentence:
    constituent_structure = constituent_parser_v2.parse(tokens_pos_tagged)
    constituency_v2_output_per_sentence.append(constituent_structure)

for tree in constituency_v2_output_per_sentence:
    print(tree) # print the sentence structure
```

```
(S
  (NP https/NN)
  :/:
  (NP
    //www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html/JJ)
  Documents/NNS
  (VP (V filed/VBN))
  to/TO
  (NP the/DT)
  (NEP San/NNP Jose/NNP)
  (NP federal/JJ court/NN)
  (P in/IN)
  (NEP California/NNP)
  (P on/IN)
  (NEP November/NNP)
  23/CD
  (NP list/NN)
  six/CD
  (NEP Samsung/NNP)
  products/NNS
  (VP (V running/VBG) (NP the/DT))
  ``/``
  Jelly/RB
  (NEP Bean/NNP)
  ''/''
  and/CC
  ``/``
  (NEP Ice/NNP Cream/NNP Sandwich/NNP)
  ''/''
  (VP (V operating/VBG))
  systems/NNS
  ,/,
  which/WDT
  (NEP Apple/NNP)
  (VP (V claims/VBZ))
  (VP (V infringe/VB))
  its/PRP$
  patents/NNS
  ./.)
(S
  (NP The/DT)
  six/CD
  phones/NNS
  and/CC
  tablets/NNS
  (VP (V affected/VBN))
  (VP (V are/VBP) (NP the/DT))
  (NEP Galaxy/NNP S/NNP III/NNP)
  ,/,
  (VP (V running/VBG) (NP the/DT new/JJ))
  (NEP Jelly/NNP Bean/NNP)
  (NP system/NN)
  ,/,
  (NP the/DT)
  (NEP Galaxy/NNP Tab/NNP)
  8.9/CD
  (NEP Wifi/NNP)
  (NP tablet/NN)
  ,/,
  (NP the/DT)
  (NEP Galaxy/NNP Tab/NNP)
  2/CD
  10.1/CD
  ,/,
  (NEP Galaxy/NNP Rugby/NNP Pro/NNP)
  and/CC
  (NEP Galaxy/NNP S/NNP III/NNP)
  (NP mini/NN)
  ./.)
(S
  (NEP Apple/NNP)
  (VP (V stated/VBD))
  it/PRP
  (VP (V had/VBD))
  (NEP “/NNP)
  (VP (V acted/VBD))
  quickly/RB
  and/CC
  diligently/RB
  ''/''
```

```
  (PP (P in/IN) (NP order/NN))
  to/TO
  ``/``
  (VP (V determine/VB) (PP (P that/IN) (NP these/DT)))
  newly/RB
  (VP (V released/VBN))
  products/NNS
  (VP (V do/VBP))
  (VP
    (V infringe/VB)
    (NP many/JJ)
    (PP (P of/IN) (NP the/DT same/JJ)))
  claims/NNS
  already/RB
  (VP (V asserted/VBN))
  (P by/IN)
  (NEP Apple/NNP)
  ./.
  ''/'')
(S
  (P In/IN)
  (NEP August/NNP)
  ,/,
  (NEP Samsung/NNP)
  (VP (V lost/VBD) (NP a/DT))
  (NEP US/NNP)
  (NP patent/NN case/NN)
  to/TO
  (NEP Apple/NNP)
  and/CC
  (VP (V was/VBD))
  (VP (V ordered/VBN))
  to/TO
  (VP (V pay/VB))
  its/PRP$
  (NP rival/JJ)
  $/$
  1.05bn/CD
  (/(
  (NP £0.66bn/NN)
  )/)
  (P in/IN)
  damages/NNS
  (P for/IN)
  (VP (V copying/VBG))
  features/NNS
  (PP (P of/IN) (NP the/DT iPad/NN))
  and/CC
  (NP iPhone/NN)
  (P in/IN)
  its/PRP$
  (NEP Galaxy/NNP)
  (NP range/NN)
  (P of/IN)
  devices/NNS
  ./.)
(S
  (NEP Samsung/NNP)
  ,/,
  which/WDT
  (VP (V is/VBZ) (NP the/DT world/NN))
  's/POS
  (NP top/JJ mobile/NN phone/NN maker/NN)
  ,/,
  (VP (V is/VBZ))
  (VP (V appealing/VBG) (NP the/DT ruling/NN))
  ./.)
(S
  (NP A/DT similar/JJ case/NN)
  (PP (P in/IN) (NP the/DT))
  (NEP UK/NNP)
  (VP (V found/VBD))
  (P in/IN)
  (NEP Samsung/NNP)
  's/POS
  (NP favour/NN)
  and/CC
  (VP (V ordered/VBD))
  (NEP Apple/NNP)
  to/TO
```

```
  (VP (V publish/VB) (NP an/DT apology/NN))
  (VP
    (V making/VBG)
    (NP clear/JJ)
    (PP (P that/IN) (NP the/DT South/JJ Korean/JJ firm/NN)))
  (VP (V had/VBD))
  not/RB
  (VP (V copied/VBN))
  its/PRP$
  (NP iPad/NN)
  when/WRB
  (VP (V designing/VBG))
  its/PRP$
  (NP own/JJ)
  devices/NNS
  ./.)
```

In [27]:
```
print(constituency_v2_output_per_sentence)
```

```
[Tree('S', [Tree('NP', [('https', 'NN')]), (':', ':'), Tree('NP', [('//www.telegraph.co.uk/technology/apple/9702716/
Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html', 'JJ')]), ('Documents', 'NNS'), Tree('VP', [Tree('V',
[('filed', 'VBN')])]), ('to', 'TO'), Tree('NP', [('the', 'DT')]), Tree('NEP', [('San', 'NNP'), ('Jose', 'NNP')]), Tr
ee('NP', [('federal', 'JJ'), ('court', 'NN')]), Tree('P', [('in', 'IN')]), Tree('NEP', [('California', 'NNP')]), Tre
e('P', [('on', 'IN')]), Tree('NEP', [('November', 'NNP')]), ('23', 'CD'), Tree('NP', [('list', 'NN')]), ('six', 'C
D'), Tree('NEP', [('Samsung', 'NNP')]), ('products', 'NNS'), Tree('VP', [Tree('V', [('running', 'VBG')]), Tree('NP',
[('the', 'DT')])]), ('``', '``'), ('Jelly', 'RB'), Tree('NEP', [('Bean', 'NNP')]), ("''", "''"), ('and', 'CC'), ('`
`', '``'), Tree('NEP', [('Ice', 'NNP'), ('Cream', 'NNP'), ('Sandwich', 'NNP')]), ("''", "''"), Tree('VP', [Tree('V',
[('operating', 'VBG')])]), ('systems', 'NNS'), (',', ','), ('which', 'WDT'), Tree('NEP', [('Apple', 'NNP')]), Tree
('VP', [Tree('V', [('claims', 'VBZ')])]), Tree('VP', [Tree('V', [('infringe', 'VB')])]), ('its', 'PRP$'), ('patent
s', 'NNS'), ('.', '.')]), Tree('S', [Tree('NP', [('The', 'DT')]), ('six', 'CD'), ('phones', 'NNS'), ('and', 'CC'),
('tablets', 'NNS'), Tree('VP', [Tree('V', [('affected', 'VBN')])]), Tree('VP', [Tree('V', [('are', 'VBP')]), Tree('N
P', [('the', 'DT')])]), Tree('NEP', [('Galaxy', 'NNP'), ('S', 'NNP'), ('III', 'NNP')]), (',', ','), Tree('VP', [Tree
('V', [('running', 'VBG')]), Tree('NP', [('the', 'DT'), ('new', 'JJ')])]), Tree('NEP', [('Jelly', 'NNP'), ('Bean',
'NNP')]), Tree('NP', [('system', 'NN')]), (',', ','), Tree('NP', [('the', 'DT')]), Tree('NEP', [('Galaxy', 'NNP'),
('Tab', 'NNP')]), ('8.9', 'CD'), Tree('NEP', [('Wifi', 'NNP')]), Tree('NP', [('tablet', 'NN')]), (',', ','), Tree('N
P', [('the', 'DT')]), Tree('NEP', [('Galaxy', 'NNP'), ('Tab', 'NNP')]), ('2', 'CD'), ('10.1', 'CD'), (',', ','), Tre
e('NEP', [('Galaxy', 'NNP'), ('Rugby', 'NNP'), ('Pro', 'NNP')]), ('and', 'CC'), Tree('NEP', [('Galaxy', 'NNP'),
('S', 'NNP'), ('III', 'NNP')]), Tree('NP', [('mini', 'NN')]), ('.', '.')]), Tree('S', [Tree('NEP', [('Apple', 'NN
P')]), Tree('VP', [Tree('V', [('stated', 'VBD')])]), ('it', 'PRP'), Tree('VP', [Tree('V', [('had', 'VBD')])]), Tree
('NEP', [('“', 'NNP')]), Tree('VP', [Tree('V', [('acted', 'VBD')])]), ('quickly', 'RB'), ('and', 'CC'), ('diligentl
y', 'RB'), ("''", "''"), Tree('PP', [Tree('P', [('in', 'IN')]), Tree('NP', [('order', 'NN')])]), ('to', 'TO'), ('`
`', '``'), Tree('VP', [Tree('V', [('determine', 'VB')]), Tree('PP', [Tree('P', [('that', 'IN')]), Tree('NP', [('thes
e', 'DT')])])]), ('newly', 'RB'), Tree('VP', [Tree('V', [('released', 'VBN')])]), ('products', 'NNS'), Tree('VP', [T
ree('V', [('do', 'VBP')])]), Tree('VP', [Tree('V', [('infringe', 'VB')]), Tree('NP', [('many', 'JJ')]), Tree('PP',
[Tree('P', [('of', 'IN')]), Tree('NP', [('the', 'DT'), ('same', 'JJ')])])]), ('claims', 'NNS'), ('already', 'RB'), T
ree('VP', [Tree('V', [('asserted', 'VBN')])]), Tree('P', [('by', 'IN')]), Tree('NEP', [('Apple', 'NNP')]), ('.',
'.'), ("''", "''")]), Tree('S', [Tree('P', [('In', 'IN')]), Tree('NEP', [('August', 'NNP')]), (',', ','), Tree('NE
P', [('Samsung', 'NNP')]), Tree('VP', [Tree('V', [('lost', 'VBD')]), Tree('NP', [('a', 'DT')])]), Tree('NEP', [('U
S', 'NNP')]), Tree('NP', [('patent', 'NN'), ('case', 'NN')]), ('to', 'TO'), Tree('NEP', [('Apple', 'NNP')]), ('and',
'CC'), Tree('VP', [Tree('V', [('was', 'VBD')])]), Tree('VP', [Tree('V', [('ordered', 'VBN')])]), ('to', 'TO'), Tree
('VP', [Tree('V', [('pay', 'VB')])]), ('its', 'PRP$'), Tree('NP', [('rival', 'JJ')]), ('$', '$'), ('1.05bn', 'CD'),
('(', '('), Tree('NP', [('£0.66bn', 'NN')]), (')', ')'), Tree('P', [('in', 'IN')]), ('damages', 'NNS'), Tree('P',
[('for', 'IN')]), Tree('VP', [Tree('V', [('copying', 'VBG')])]), ('features', 'NNS'), Tree('PP', [Tree('P', [('of',
'IN')]), Tree('NP', [('the', 'DT'), ('iPad', 'NN')])]), ('and', 'CC'), Tree('NP', [('iPhone', 'NN')]), Tree('P',
[('in', 'IN')]), ('its', 'PRP$'), Tree('NEP', [('Galaxy', 'NNP')]), Tree('NP', [('range', 'NN')]), Tree('P', [('of',
'IN')]), ('devices', 'NNS'), ('.', '.')]), Tree('S', [Tree('NEP', [('Samsung', 'NNP')]), (',', ','), ('which', 'WD
T'), Tree('VP', [Tree('V', [('is', 'VBZ')]), Tree('NP', [('the', 'DT'), ('world', 'NN')])]), ("'s", 'POS'), Tree('N
P', [('top', 'JJ'), ('mobile', 'NN'), ('phone', 'NN'), ('maker', 'NN')]), (',', ','), Tree('VP', [Tree('V', [('is',
'VBZ')])]), Tree('VP', [Tree('V', [('appealing', 'VBG')]), Tree('NP', [('the', 'DT'), ('ruling', 'NN')])]), ('.',
'.')]), Tree('S', [Tree('NP', [('A', 'DT'), ('similar', 'JJ'), ('case', 'NN')]), Tree('PP', [Tree('P', [('in', 'I
N')]), Tree('NP', [('the', 'DT')])]), Tree('NEP', [('UK', 'NNP')]), Tree('VP', [Tree('V', [('found', 'VBD')])]), Tre
e('P', [('in', 'IN')]), Tree('NEP', [('Samsung', 'NNP')]), ("'s", 'POS'), Tree('NP', [('favour', 'NN')]), ('and', 'C
C'), Tree('VP', [Tree('V', [('ordered', 'VBD')])]), Tree('NEP', [('Apple', 'NNP')]), ('to', 'TO'), Tree('VP', [Tree
('V', [('publish', 'VB')]), Tree('NP', [('an', 'DT'), ('apology', 'NN')])]), Tree('VP', [Tree('V', [('making', 'VB
G')]), Tree('NP', [('clear', 'JJ')]), Tree('PP', [Tree('P', [('that', 'IN')]), Tree('NP', [('the', 'DT'), ('South',
'JJ'), ('Korean', 'JJ'), ('firm', 'NN')])])]), Tree('VP', [Tree('V', [('had', 'VBD')])]), ('not', 'RB'), Tree('VP',
[Tree('V', [('copied', 'VBN')])]), ('its', 'PRP$'), Tree('NP', [('iPad', 'NN')]), ('when', 'WRB'), Tree('VP', [Tree
('V', [('designing', 'VBG')])]), ('its', 'PRP$'), Tree('NP', [('own', 'JJ')]), ('devices', 'NNS'), ('.', '.')])]
```

[total points: 1] Exercise 2: spaCy

Use Spacy to process the same text as you analyzed with NLTK.

In [28]:
```
import spacy
try:
    nlp = spacy.load('en_core_web_sm')
except OSError:
    !python -m spacy download en_core_web_sm
    nlp = spacy.load('en_core_web_sm')
```

In [29]:
```
doc = nlp(text)
sents = list(doc.sents)
```

small tip: You can use **sents = list(doc.sents)** to be able to use the index to access a sentence like **sents[2]** for the third sentence.

In [30]:
```
# Sentence splitting & tokenization
for sent in sents:
    print("")
    print(sent)
    for token in sent:
        print(token.text)
```

https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html

Documents filed to the San Jose federal court in California on November 23 list six Samsung products running the "Jelly Bean" and "Ice Cream Sandwich" operating systems, which Apple claims infringe its patents.

https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html

Documents
filed
to
the
San
Jose
federal
court
in
California
on
November
23
list
six
Samsung
products
running
the
"
Jelly
Bean
"
and
"
Ice
Cream
Sandwich
"
operating
systems
,
which
Apple
claims
infringe
its
patents
.

The six phones and tablets affected are the Galaxy S III, running the new Jelly Bean system, the Galaxy Tab 8.9 Wifi tablet, the Galaxy Tab 2 10.1, Galaxy Rugby Pro and Galaxy S III mini.

The
six
phones
and
tablets
affected
are
the
Galaxy
S
III
,
running
the
new
Jelly
Bean
system
,
the
Galaxy
Tab
8.9
Wifi
tablet
,

the
Galaxy
Tab
2
10.1
,
Galaxy
Rugby
Pro
and
Galaxy
S
III
mini
.

Apple stated it had “acted quickly and diligently" in order to "determine that these newly released products do infringe many of the same claims already asserted by Apple.

Apple
stated
it
had
“
acted
quickly
and
diligently
"
in
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to
"
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that
these
newly
released
products
do
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many
of
the
same
claims
already
asserted
by
Apple
.

"
In August, Samsung lost a US patent case to Apple and was ordered to pay its rival $1.05bn (£0.66bn) in damages for copying features of the iPad and iPhone in its Galaxy range of devices.
"

In
August
,
Samsung
lost
a
US
patent
case
to
Apple
and
was
ordered
to
pay
its
rival
$
1.05bn
(

£
0.66bn
)
in
damages
for
copying
features
of
the
iPad
and
iPhone
in
its
Galaxy
range
of
devices
.

Samsung, which is the world's top mobile phone maker, is appealing the ruling.

Samsung
,
which
is
the
world
's
top
mobile
phone
maker
,
is
appealing
the
ruling
.

A similar case in the UK found in Samsung's favour and ordered Apple to publish an apology making clear that the South Korean firm had not copied its iPad when designing its own devices.
A
similar
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in
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's
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apology
making
clear
that
the
South
Korean
firm
had
not
copied
its
iPad
when
designing
its
own

```
devices
.
```

In [31]:

```python
# Part of speech tagging
for sent in sents:
    print("")
    for token in sent:
        print(token.text, token.pos_, token.tag_)
```

https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html PRO
PN NNP

SPACE _SP
Documents PROPN NNPS
filed VERB VBD
to ADP IN
the DET DT
San PROPN NNP
Jose PROPN NNP
federal ADJ JJ
court NOUN NN
in ADP IN
California PROPN NNP
on ADP IN
November PROPN NNP
23 NUM CD
list NOUN NN
six NUM CD
Samsung PROPN NNP
products NOUN NNS
running VERB VBG
the DET DT
" PUNCT ``
Jelly PROPN NNP
Bean PROPN NNP
" PUNCT ''
and CCONJ CC
" PUNCT ``
Ice PROPN NNP
Cream PROPN NNP
Sandwich PROPN NNP
" PUNCT ''
operating NOUN NN
systems NOUN NNS
, PUNCT ,
which PRON WDT
Apple PROPN NNP
claims VERB VBZ
infringe VERB VBP
its PRON PRP$
patents NOUN NNS
. PUNCT .

SPACE _SP

The DET DT
six NUM CD
phones NOUN NNS
and CCONJ CC
tablets NOUN NNS
affected VERB VBN
are AUX VBP
the DET DT
Galaxy PROPN NNP
S PROPN NNP
III PROPN NNP
, PUNCT ,
running VERB VBG
the DET DT
new ADJ JJ
Jelly PROPN NNP
Bean PROPN NNP
system NOUN NN
, PUNCT ,
the DET DT
Galaxy PROPN NNP
Tab PROPN NNP
8.9 NUM CD
Wifi PROPN NNP
tablet NOUN NN
, PUNCT ,
the DET DT
Galaxy PROPN NNP
Tab PROPN NNP
2 NUM CD
10.1 NUM CD
, PUNCT ,
Galaxy PROPN NNP

Rugby PROPN NNP
Pro PROPN NNP
and CCONJ CC
Galaxy PROPN NNP
S PROPN NNP
III PROPN NNP
mini NOUN NN
. PUNCT .

 SPACE _SP

Apple PROPN NNP
stated VERB VBD
it PRON PRP
had AUX VBD
“ PUNCT ``
acted VERB VBN
quickly ADV RB
and CCONJ CC
diligently ADV RB
" PUNCT ''
in ADP IN
order NOUN NN
to PART TO
" PUNCT ``
determine VERB VB
that SCONJ IN
these DET DT
newly ADV RB
released VERB VBN
products NOUN NNS
do AUX VBP
infringe VERB VB
many ADJ JJ
of ADP IN
the DET DT
same ADJ JJ
claims NOUN NNS
already ADV RB
asserted VERB VBN
by ADP IN
Apple PROPN NNP
. PUNCT .

" PUNCT ''

 SPACE _SP
In ADP IN
August PROPN NNP
, PUNCT ,
Samsung PROPN NNP
lost VERB VBD
a DET DT
US PROPN NNP
patent NOUN NN
case NOUN NN
to ADP IN
Apple PROPN NNP
and CCONJ CC
was AUX VBD
ordered VERB VBN
to PART TO
pay VERB VB
its PRON PRP$
rival ADJ JJ
$ SYM $
1.05bn NUM CD
(PUNCT -LRB-
£ SYM $
0.66bn NUM CD
) PUNCT -RRB-
in ADP IN
damages NOUN NNS
for ADP IN
copying VERB VBG
features NOUN NNS
of ADP IN
the DET DT
iPad PROPN NNP
and CCONJ CC

```
iPhone PROPN NNP
in ADP IN
its PRON PRP$
Galaxy PROPN NNP
range NOUN NN
of ADP IN
devices NOUN NNS
. PUNCT .

Samsung PROPN NNP
, PUNCT ,
which PRON WDT
is AUX VBZ
the DET DT
world NOUN NN
's PART POS
top ADJ JJ
mobile ADJ JJ
phone NOUN NN
maker NOUN NN
, PUNCT ,
is AUX VBZ
appealing VERB VBG
the DET DT
ruling NOUN NN
. PUNCT .

 SPACE _SP

A DET DT
similar ADJ JJ
case NOUN NN
in ADP IN
the DET DT
UK PROPN NNP
found VERB VBD
in ADP IN
Samsung PROPN NNP
's PART POS
favour NOUN NN
and CCONJ CC
ordered VERB VBD
Apple PROPN NNP
to PART TO
publish VERB VB
an DET DT
apology NOUN NN
making NOUN NN
clear ADJ JJ
that SCONJ IN
the DET DT
South ADJ JJ
Korean ADJ JJ
firm NOUN NN
had AUX VBD
not PART RB
copied VERB VBN
its PRON PRP$
iPad PROPN NNP
when SCONJ WRB
designing VERB VBG
its PRON PRP$
own ADJ JJ
devices NOUN NNS
. PUNCT .
```

In [32]:

```python
# Named Entity Recognition (NER)
for ent in doc.ents:
    print(ent.text, ent.label_)
```

```
San Jose GPE
California GPE
November 23 DATE
six CARDINAL
Samsung ORG
Jelly Bean WORK_OF_ART
Apple ORG
six CARDINAL
the Galaxy S III PERSON
Jelly Bean ORG
Galaxy Tab 8.9 Wifi PERSON
Galaxy Tab 2 10.1 PERSON
Apple ORG
Apple ORG
August DATE
Samsung ORG
US GPE
Apple ORG
1.05bn MONEY
0.66bn MONEY
iPad ORG
iPhone ORG
Samsung ORG
UK GPE
Samsung ORG
Apple ORG
South Korean NORP
iPad ORG
```

In [33]:
```python
# NER visualization (from Lab1.3)
from spacy import displacy
displacy.render(doc, jupyter=True, style='ent')
```

https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html

Documents filed to the San Jose GPE federal court in California GPE on November 23 DATE list six CARDINAL Samsung ORG products running the " Jelly Bean WORK_OF_ART " and "Ice Cream Sandwich" operating systems, which Apple ORG claims infringe its patents.

The six CARDINAL phones and tablets affected are the Galaxy S III PERSON , running the new Jelly Bean ORG system, the Galaxy Tab 8.9 Wifi PERSON tablet, the Galaxy Tab 2 10.1 PERSON , Galaxy Rugby Pro and Galaxy S III mini.

Apple ORG stated it had “acted quickly and diligently" in order to "determine that these newly released products do infringe many of the same claims already asserted by Apple ORG ."

In August DATE , Samsung ORG lost a US GPE patent case to Apple ORG and was ordered to pay its rival $ 1.05bn MONEY (£ 0.66bn MONEY) in damages for copying features of the iPad ORG and iPhone ORG in its Galaxy range of devices.

Samsung ORG , which is the world's top mobile phone maker, is appealing the ruling.

A similar case in the UK GPE found in Samsung ORG 's favour and ordered Apple ORG to publish an apology making clear that the South Korean NORP firm had not copied its iPad ORG when designing its own devices.

In [34]:
```python
# Dependency parsing
for sent in sents:
    print("")
    for token in sent:
        print(token.text, token.dep_, token.head)
```

https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.html com
pound Documents

dep https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.htm
l
Documents nsubj filed
filed ROOT filed
to prep filed
the det court
San compound Jose
Jose nmod court
federal amod court
court pobj to
in prep court
California pobj in
on prep filed
November pobj on
23 nummod November
list npadvmod filed
six nummod products
Samsung compound products
products appos list
running acl products
the det Bean
" punct Bean
Jelly compound Bean
Bean dobj running
" punct Bean
and cc Bean
" punct Sandwich
Ice compound Cream
Cream compound Sandwich
Sandwich nmod systems
" punct Sandwich
operating compound systems
systems conj Bean
, punct systems
which nsubj claims
Apple nsubj claims
claims relcl systems
infringe xcomp claims
its poss patents
patents dobj infringe
. punct filed

dep .

The det phones
six nummod phones
phones nsubj are
and cc phones
tablets conj phones
affected acl phones
are ROOT are
the det III
Galaxy compound III
S compound III
III attr are
, punct are
running advcl are
the det system
new amod system
Jelly compound Bean
Bean compound system
system dobj running
, punct running
the det tablet
Galaxy compound Tab
Tab nmod tablet
8.9 nummod Tab
Wifi compound tablet
tablet dep are
, punct tablet
the det Tab
Galaxy compound Tab
Tab appos tablet
2 nummod 10.1
10.1 appos Tab
, punct tablet

```
Galaxy compound Pro
Rugby compound Pro
Pro appos tablet
and cc Pro
Galaxy compound S
S compound III
III conj Pro
mini conj Pro
. punct are

 dep .

Apple nsubj stated
stated ROOT stated
it nsubj acted
had aux acted
“ punct acted
acted ccomp stated
quickly advmod acted
and cc quickly
diligently conj quickly
" punct acted
in prep acted
order pobj in
to aux determine
" punct determine
determine acl order
that mark infringe
these det products
newly advmod released
released amod products
products nsubj infringe
do aux infringe
infringe ccomp determine
many dobj infringe
of prep many
the det claims
same amod claims
claims pobj of
already advmod asserted
asserted acl claims
by agent asserted
Apple pobj by
. punct stated

" punct lost

 dep "
In prep lost
August pobj In
, punct lost
Samsung nsubj lost
lost ROOT lost
a det case
US compound case
patent compound case
case dobj lost
to prep lost
Apple pobj to
and cc lost
was auxpass ordered
ordered conj lost
to aux pay
pay xcomp ordered
its poss rival
rival dobj pay
$ nmod 1.05bn
1.05bn appos rival
( punct 1.05bn
£ nmod 0.66bn
0.66bn appos 1.05bn
) punct 1.05bn
in prep pay
damages pobj in
for prep damages
copying pcomp for
features dobj copying
of prep features
the det iPad
iPad pobj of
```

```
and cc iPad
iPhone conj iPad
in prep copying
its poss range
Galaxy compound range
range pobj in
of prep range
devices pobj of
. punct lost

Samsung nsubj appealing
, punct Samsung
which nsubj is
is relcl Samsung
the det world
world poss maker
's case world
top amod maker
mobile amod phone
phone compound maker
maker attr is
, punct Samsung
is aux appealing
appealing ROOT appealing
the det ruling
ruling dobj appealing
. punct appealing

 dep .

A det case
similar amod case
case ROOT case
in prep case
the det UK
UK pobj in
found acl case
in prep found
Samsung poss favour
's case Samsung
favour pobj in
and cc found
ordered conj found
Apple dobj ordered
to aux publish
publish xcomp ordered
an det making
apology compound making
making dobj publish
clear amod making
that mark copied
the det firm
South amod Korean
Korean amod firm
firm nsubj copied
had aux copied
not neg copied
copied ccomp ordered
its poss iPad
iPad dobj copied
when advmod designing
designing advcl copied
its poss devices
own amod devices
devices dobj designing
. punct case
```

[total points: 7] Exercise 3: Comparison NLTK and spaCy

We will now compare the output of NLTK and spaCy, i.e., in what do they differ?

[points: 3] Exercise 3a: Part of speech tagging

Compare the output from NLTK and spaCy regarding part of speech tagging.

- To compare, you probably would like to compare sentence per sentence. Describe if the sentence splitting is different for NLTK than for spaCy. If not, where do they differ?

- After checking the sentence splitting, select a sentence for which you expect interesting results and perhaps differences. Motivate your choice.
- Compare the output in `token.tag` from spaCy to the part of speech tagging from NLTK for each token in your selected sentence. Are there any differences? This is not a trick question; it is possible that there are no differences.

In [35]:
```
# Sentence splitting comparison
print(f"Number of NLTK sentences: {len(sentences_nltk)}")
print(f"Number of spaCy sentences: {len(sents)}")

print("\nNLTK sentences:")
for i, s in enumerate(sentences_nltk):
    print(f"  [{i}]: {s}")

print("\nspaCy sentences:")
for i, s in enumerate(sents):
    clean = s.text.strip()
    print(f"  [{i}]: {clean}")

# POS comparison for "The six phones and tablets affected..." sentence
target = "The six phones"
nltk_idx = next(i for i, s in enumerate(sentences_nltk) if target in s)
spacy_idx = next(i for i, s in enumerate(sents) if target in s.text)

print(f"\n=== POS Tag Comparison for sentence (NLTK idx={nltk_idx}, spaCy idx={spacy_idx}) ===\n")

spacy_tokens = [(t.text, t.tag_) for t in sents[spacy_idx] if t.text.strip()]
nltk_tokens = pos_tags_per_sentence[nltk_idx]

print(f"{'Token':<15} {'NLTK':<10} {'spaCy':<10} {'Match'}")
print("-" * 45)
differences = 0
for i in range(min(len(nltk_tokens), len(spacy_tokens))):
    nt, ntag = nltk_tokens[i]
    st, stag = spacy_tokens[i]
    match = "Yes" if ntag == stag else "NO  <--"
    if ntag != stag:
        differences += 1
    print(f"{nt:<15} {ntag:<10} {stag:<10} {match}")

print(f"\nTotal differences: {differences}")
```

```
Number of NLTK sentences: 6
Number of spaCy sentences: 6

NLTK sentences:
  [0]: https://www.telegraph.co.uk/technology/apple/9702716/Apple-Samsung-lawsuit-six-more-products-under-scrutiny.h
tml

Documents filed to the San Jose federal court in California on November 23 list six Samsung products running the "Je
lly Bean" and "Ice Cream Sandwich" operating systems, which Apple claims infringe its patents.
  [1]: The six phones and tablets affected are the Galaxy S III, running the new Jelly Bean system, the Galaxy Tab
8.9 Wifi tablet, the Galaxy Tab 2 10.1, Galaxy Rugby Pro and Galaxy S III mini.
  [2]: Apple stated it had “acted quickly and diligently" in order to "determine that these newly released products
do infringe many of the same claims already asserted by Apple."
  [3]: In August, Samsung lost a US patent case to Apple and was ordered to pay its rival $1.05bn (£0.66bn) in damag
es for copying features of the iPad and iPhone in its Galaxy range of devices.
  [4]: Samsung, which is the world's top mobile phone maker, is appealing the ruling.
  [5]: A similar case in the UK found in Samsung's favour and ordered Apple to publish an apology making clear that
the South Korean firm had not copied its iPad when designing its own devices.

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the South Korean firm had not copied its iPad when designing its own devices.

=== POS Tag Comparison for sentence (NLTK idx=1, spaCy idx=1) ===

Token           NLTK       spaCy      Match
----------------------------------------------
The             DT         DT         Yes
six             CD         CD         Yes
phones          NNS        NNS        Yes
and             CC         CC         Yes
tablets         NNS        NNS        Yes
affected        VBN        VBN        Yes
are             VBP        VBP        Yes
the             DT         DT         Yes
Galaxy          NNP        NNP        Yes
S               NNP        NNP        Yes
III             NNP        NNP        Yes
,               ,          ,          Yes
running         VBG        VBG        Yes
the             DT         DT         Yes
new             JJ         JJ         Yes
Jelly           NNP        NNP        Yes
Bean            NNP        NNP        Yes
system          NN         NN         Yes
,               ,          ,          Yes
the             DT         DT         Yes
Galaxy          NNP        NNP        Yes
Tab             NNP        NNP        Yes
8.9             CD         CD         Yes
Wifi            NNP        NNP        Yes
tablet          NN         NN         Yes
,               ,          ,          Yes
the             DT         DT         Yes
Galaxy          NNP        NNP        Yes
Tab             NNP        NNP        Yes
2               CD         CD         Yes
10.1            CD         CD         Yes
,               ,          ,          Yes
Galaxy          NNP        NNP        Yes
Rugby           NNP        NNP        Yes
Pro             NNP        NNP        Yes
and             CC         CC         Yes
Galaxy          NNP        NNP        Yes
S               NNP        NNP        Yes
III             NNP        NNP        Yes
mini            NN         NN         Yes
```

```
.                .          .          Yes

Total differences: 0
```

Sentence splitting:

Both NLTK and spaCy produce exactly **6 sentences**, and the sentence boundaries are nearly identical. The only difference is in **quote handling** around sentence 2/3. Sentence 2 ends with `Apple."`, the period is inside the closing quotation mark. NLTK recognizes `."` as a sentence ending and attaches the closing quote to sentence 2, so sentence 3 cleanly starts with `In August...`. spaCy, however, splits the period and quotation mark into separate tokens and does not attach the closing `"` to sentence 2. Instead, that orphaned `"` gets pushed to the start of sentence 3, resulting in `" In August, Samsung lost...`. This shows that spaCy's sentence splitter struggles with punctuation sequences like `."` at sentence boundaries.

Both tools merge the URL with the "Documents filed..." paragraph into a single sentence (sentence 0). This is because sentence tokenizers rely on punctuation (`.`, `?`, `!`) to detect boundaries: the URL ending in `.html` is not recognized as a sentence-ending period, and the blank line (newline characters) between the URL and the paragraph is not treated as a sentence boundary either. This is a limitation when processing text that contains non-natural-language elements like URLs.

Sentence choice:

We selected sentence index 1 ("The six phones and tablets affected are the Galaxy S III...") because it contains many product names (Galaxy S III, Jelly Bean, Galaxy Tab), version numbers (8.9, 10.1), and a mix of proper nouns, common nouns, and adjectives, making it a good candidate for revealing POS tagging differences.

POS tag comparison:

For the selected sentence, the fine-grained POS tags (`tag_`) from both NLTK and spaCy are **identical** for all 41 tokens (0 differences). Both tools use the Penn Treebank tagset and fully agree on this sentence.

However, differences do appear in other sentences. For sentence 0 ("Documents filed to the San Jose federal court..."), where the URL causes different tokenization, comparing the shared tokens reveals:

- **"Jelly"**: NLTK → RB (adverb), spaCy → NNP (proper noun). spaCy is correct here since "Jelly Bean" is a proper noun (Android OS version name).
- **"Documents"**: NLTK → NNS (common plural noun), spaCy → NNPS (proper plural noun).
- **"filed"**: NLTK → VBN (past participle), spaCy → VBD (simple past). NLTK arguably captures the reduced relative clause ("Documents [that were] filed") better.
- **"operating"**: NLTK → VBG (gerund), spaCy → NN (noun). In "operating systems," spaCy treats it as part of a compound noun.

These differences show that while both tools use the same tagset, their underlying models can disagree on ambiguous words.

[points: 2] Exercise 3b: Named Entity Recognition (NER)

- Describe differences between the output from NLTK and spaCy for Named Entity Recognition. Which one do you think performs better?

Differences between NLTK and spaCy NER:

1. **Entity type richness**: spaCy uses a much richer set of entity labels: ORG, GPE, DATE, MONEY, CARDINAL, NORP (nationality/religious/political group), and WORK_OF_ART. NLTK uses a smaller set: ORGANIZATION, PERSON, GPE, and GSP.

2. **Multi-word entities**: spaCy correctly groups multi-word entities (e.g., "November 23" as DATE, "Galaxy Tab 8.9 Wifi" as one entity, "South Korean" as NORP). NLTK typically recognizes only individual words or short spans (e.g., "Samsung" alone, "Galaxy" alone).

3. **Numeric/value entities**: spaCy recognizes numbers ("six" as CARDINAL), dates ("August" as DATE), and monetary values ("$1.05bn" as MONEY, "£0.66bn" as MONEY). NLTK misses almost all of these.

4. **Misclassifications**:

   - NLTK labels "Apple" as PERSON in some sentences instead of ORGANIZATION.
   - NLTK labels "Bean" (from "Jelly Bean") as GPE (a location).
   - NLTK labels "August" as GPE.
   - spaCy mislabels product names like "the Galaxy S III" and "Galaxy Tab 8.9 Wifi" as PERSON instead of a product entity.
   - spaCy labels "Jelly Bean" as WORK_OF_ART in one instance and ORG in another: inconsistent.

Which performs better?

spaCy performs significantly better overall. It recognizes more entities, groups multi-word entities correctly, uses a richer label set, and makes fewer critical misclassifications. While it incorrectly labels some Samsung product names as PERSON, its overall accuracy and coverage far exceed NLTK's rule-based NER module.

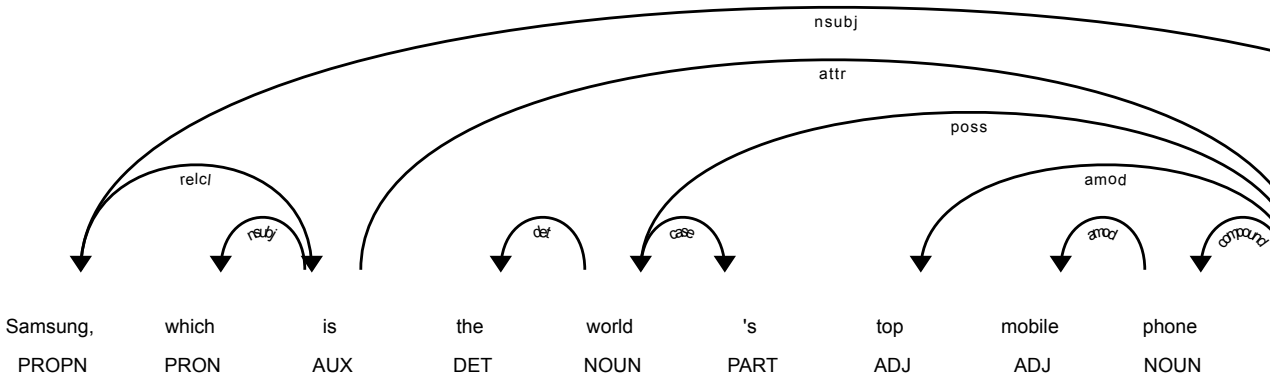
[points: 2] Exercise 3c: Constituency/dependency parsing

Choose one sentence from the text and run constituency parsing using NLTK and dependency parsing using spaCy.

- describe briefly the difference between constituency parsing and dependency parsing
- describe differences between the output from NLTK and spaCy.

In [43]:
```
# Choose "Samsung, which is the world's top mobile phone maker, is appealing the ruling."
target = "world"
nltk_idx = next(i for i, s in enumerate(sentences_nltk) if target in s)
spacy_idx = next(i for i, s in enumerate(sents) if target in s.text)

print("Selected sentence:", sentences_nltk[nltk_idx].strip())

print("\n=== NLTK Constituency Parsing (v2, with NEP rule) ===\n")
print(constituency_v2_output_per_sentence[nltk_idx])

print("\n=== spaCy Dependency Parsing ===\n")
print(f"{'Token':<15} {'Dep Label':<12} {'Explanation':<30} {'Head'}")
print("-" * 70)
for token in sents[spacy_idx]:
    if token.text.strip():
        explanation = spacy.explain(token.dep_) or ""
        print(f"{token.text:<15} {token.dep_:<12} {explanation:<30} {token.head.text}")

print("\n=== spaCy Dependency Tree Visualization ===")
displacy.render(sents[spacy_idx], jupyter=True, style='dep', options={"distance": 100})
```

```
Selected sentence: Samsung, which is the world's top mobile phone maker, is appealing the ruling.

=== NLTK Constituency Parsing (v2, with NEP rule) ===

(S
  (NEP Samsung/NNP)
  ,/,
  which/WDT
  (VP (V is/VBZ) (NP the/DT world/NN))
  's/POS
  (NP top/JJ mobile/NN phone/NN maker/NN)
  ,/,
  (VP (V is/VBZ))
  (VP (V appealing/VBG) (NP the/DT ruling/NN))
  ./.)

=== spaCy Dependency Parsing ===

Token           Dep Label    Explanation                    Head
----------------------------------------------------------------------
Samsung         nsubj        nominal subject                appealing
,               punct        punctuation                    Samsung
which           nsubj        nominal subject                is
is              relcl        relative clause modifier       Samsung
the             det          determiner                     world
world           poss         possession modifier            maker
's              case         case marking                   world
top             amod         adjectival modifier            maker
mobile          amod         adjectival modifier            phone
phone           compound     compound                       maker
maker           attr         attribute                      is
,               punct        punctuation                    Samsung
is              aux          auxiliary                      appealing
appealing       ROOT         root                           appealing
the             det          determiner                     ruling
ruling          dobj         direct object                  appealing
.               punct        punctuation                    appealing

=== spaCy Dependency Tree Visualization ===
```



Constituency parsing vs. dependency parsing:

- **Constituency parsing** decomposes a sentence into nested sub-phrases (constituents) in a tree structure. Each internal node represents a phrase type (NP, VP, PP), and the leaves are individual words. It answers: *"How do words group into phrases?"*
- **Dependency parsing** identifies directed grammatical relationships between individual words. Each word (dependent) is connected to exactly one other word (head) via a labeled relation (e.g., nsubj, dobj, amod). It answers: *"Which word depends on which other word?"*

Output differences for the selected sentence:

NLTK's `RegexpParser` (v2, with our added NEP rule `{<NNP>+}`) produces a rule-based constituency tree. It captures phrase structures that match the regex patterns (NP, VP, PP, NEP), and the NEP rule successfully groups "Samsung" as a named entity phrase. However, many tokens are still left unchunked directly under the root `S` node — function words like "which" and the possessive "'s" are not grouped into any phrase. The output remains incomplete and shallow.

spaCy's dependency parser produces a fully connected graph where *every* token has an assigned head and dependency label. Using `spacy.explain()` (as taught in Lab1.3), we can interpret each label. It correctly identifies:

- "Samsung" as the subject (nsubj) of "appealing"
- "which" as subject (nsubj) of "is" in the relative clause
- "maker" as the attribute (attr) of the copular "is"
- "ruling" as the direct object (dobj) of "appealing"
- Possessive structure: "'s" marks "world" as the possessor of "maker"

The `displacy.render()` visualization (also from Lab1.3) makes the dependency structure much easier to read than the raw text output, clearly showing the directed arcs between tokens.

The spaCy output is more informative and accurate because it uses a trained statistical model that handles complex constructions (relative clauses, possessives, appositives), while NLTK's rule-based approach is limited to what the handwritten regex rules can capture.

End of this notebook